Experiment 1: Why the Estimators Err the Way They Do

Insight analysis for Chapter 4, Section 4.1

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Purpose of this document. The raw write-up of Experiment 1 (`insights/experiment_1/`) reports how far each estimator is from the truth on the five static recordings. This document asks what kind of error each one makes and when it is made: whether wheel odometry's error is a scale or a heading error, why IMU dead reckoning diverges, when and why the EKF's covariance stops describing its error, and what the visual estimators' error consists of once they are scored at the IMU point. Each question is tested within every recording, so the recording is its own control, and the answer carries the evidence for and against it. Every number is written by `script/exp1_insights.py` into `output/compare/simulation/exp1-insights/`; the tables are input from there unchanged.

1 Questions and method

Six questions were put to the five `dataset_allsensor` recordings (replay at $0.5\times$, four worlds, one run per estimator per recording).

- Q1. Is wheel odometry's error a distance (scale) error, a heading error, or heading error carried into position?
- Q2. Does each estimator's error grow faster while the robot turns than on straights?
- Q3. Does IMU dead reckoning diverge through its heading or through its velocity?
- Q4. When does the EKF's covariance stop describing its error, and does the per-recording noise calibration change that?
- Q5. What does the transferred noise model (the VSLAM file's IMU constants) do to the EKF, mechanically?
- Q6. For the visual estimators, scored at the IMU point: what is their error made of, does it grow at turns or with few features, and what was the one failure?

Baselines. The wheel, IMU and EKF estimates start from the ground-truth pose and are compared to the raw model-origin ground truth, as in the raw write-up. The error vector is rotated into the ground-truth path frame, giving an along-track part (distance travelled) and a cross-track part (lateral), and the heading error is the estimate's yaw against the truth's, zeroed at the start. To separate heading from velocity error, the true velocity is integrated through the heading error alone; the ratio of that prediction to the actual error says how much of the error the heading explains. Growth rates while turning (ground-truth yaw rate above $10\,^\circ\,\mathrm{s}^{-1}$) and on straights are compared as in the Experiment 3 insight analysis. NEES is computed from the logged covariance at 1 Hz; the filter "leaves its band" at the first time it stays above the 97.5 % value of a 3-DOF χ^2 for 5 s.

Visual estimators. VINS-Fusion and ORB-SLAM3 report the IMU, 0.338 m ahead of the model origin, so they are scored against the ground truth moved to that point (the lever-arm finding of the Experiment 3

insight analysis, now applied in every scoring script). The initialisation transient is cut and the start-anchored error, tracking regime and turning contrast of the Experiment 3 analysis are reused. Whole-run ATE is reported both in three dimensions, as the chapter does, and in the plane.

2 Insight 1: wheel odometry's error is heading error made in the turns, not a distance error

Wheel odometry measures distance to within one percent; its position error is heading error carried forward, and the heading error is made almost entirely while the robot turns. Tables I.1 and I.2 and figure I.1 carry the evidence.

Table I.1: Anchored error of the proprioceptive baselines in the ground-truth path frame: along-track (distance) and cross-track (heading) RMS and final values, the heading error itself, the path-length (scale) error of the estimate, and the share of the final position error that the heading error alone reproduces when the true velocity is integrated through it. Model-origin reference. Source: `output/compare/simulation/exp1-insights/decomposition.csv`.

Recording	Estimator	RMS [m]	along RMS [m]	cross RMS [m]	final [m]	heading RMS [°]	scale [%]	heading-only share of final
`allsensor_000`	Wheel odometry	4.73	3.42	3.27	4.77	11.07	0.1	0.97
	IMU dead reckoning	39.35	19.92	33.94	47.60	1.00	55.8	0.01
	EKF wheel+IMU	0.28	0.24	0.15	0.12	0.35	0.7	1.73
	EKF @ VSLAM noise	1.98	1.60	1.17	1.44	8.59	22.2	2.29
`allsensor_001`	Wheel odometry	1.83	1.41	1.16	2.43	6.90	-0.2	1.04
	IMU dead reckoning	14.73	7.14	12.88	23.74	0.59	-29.3	0.01
	EKF wheel+IMU	0.35	0.31	0.17	0.30	0.42	0.1	0.49
	EKF @ VSLAM noise	0.94	0.81	0.47	1.03	4.25	16.0	1.72
`allsensor_002`	Wheel odometry	2.89	2.29	1.76	4.33	15.18	0.2	0.98
	IMU dead reckoning	16.36	8.25	14.13	17.73	0.61	1.1	0.01
	EKF wheel+IMU	0.35	0.28	0.20	0.35	0.29	0.4	0.66
	EKF @ VSLAM noise	1.32	1.12	0.71	1.80	9.05	9.0	1.55
`allsensor_003`	Wheel odometry	3.98	3.10	2.50	9.23	15.63	0.8	0.95
	IMU dead reckoning	62.90	35.36	52.03	74.75	3.20	248.7	0.01
	EKF wheel+IMU	0.67	0.58	0.34	1.24	3.54	0.7	0.75
	EKF @ VSLAM noise	1.57	1.37	0.78	3.04	9.87	11.7	1.65
`allsensor_004`	Wheel odometry	5.12	3.89	3.33	5.79	15.00	0.1	0.96
	IMU dead reckoning	41.48	21.38	35.54	48.10	0.84	102.5	0.00
	EKF wheel+IMU	0.78	0.64	0.44	0.83	3.41	0.8	1.14
	EKF @ VSLAM noise	2.65	2.22	1.44	1.84	12.62	17.8	2.55

The path length the wheels report is within 0.8 % of the true one on every recording ($n = 5$), while the heading error reaches 7–16° RMS. Integrating the true velocity through that heading error alone reproduces 95–104 % of the final position error on all five recordings. The along-track component is not a scale error: it appears once the heading error has rotated the path, which is why it is as large as the cross-track component.

The heading error is made in the turns. Its growth rate while turning is 650–1406 $\mathrm{m}^\circ\,\mathrm{s}^{-1}$ against a negative or near-zero rate on straights, in five of five recordings; the heading-error traces of figure I.1 step at each grey band and hold between them. The size of the step varies between turns and between recordings, which is why the raw write-up found no relation between the fitted yaw-rate scale and the total drift (its H1); the mechanism is nonetheless a per-turn heading error, not a continuous drift.

Table I.2: Growth rate of the position error and of the heading error while the ground-truth yaw rate exceeds $10\,^\circ\,s^{-1}$ and on straights, median over the five recordings, with the number of recordings in which the turning rate is the larger ($n_>$). Baselines: anchored error over the whole run. Visual estimators: start-anchored error at the IMU point, tracking regime only (before the first jump, below 3 m); heading contrast not computed for them.

Estimator	n	position [mm s^{-1}]			heading [m° s^{-1}]		
		turning	straight	$n_>$	turning	straight	$n_>$
Wheel odometry	5	63.6	48.2	2	845.5	-78.0	5
IMU dead reckoning	5	175.1	605.8	0	18.2	15.6	4
EKF wheel+IMU	5	7.8	2.6	4	15.3	6.7	3
EKF @ VSLAM noise	5	76.1	-11.0	5	538.6	-13.4	5
VINS-Fusion	5	0.4	5.4	1	—	—	—
ORB-SLAM3	4	12.3	0.3	3	—	—	—
ORB-SLAM3 (no LC)	4	-1.7	1.8	1	—	—	—

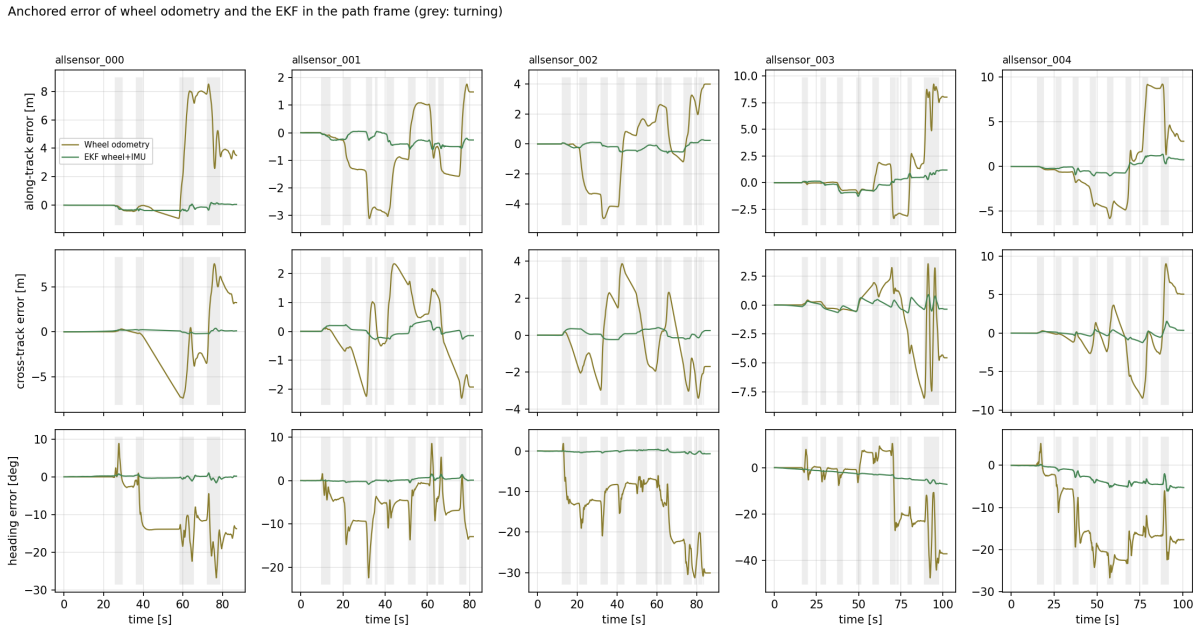


Figure I.1: Anchored error of wheel odometry and the EKF on each recording, in the ground-truth path frame: along-track error, cross-track error and heading error against time. Grey bands are the moments the ground-truth yaw rate exceeds $10\,^\circ\,s^{-1}$.

3 Insight 2: IMU dead reckoning diverges through its velocity, not its heading

IMU-only dead reckoning keeps its heading to within a few degrees and loses its position through the velocity integral, whose error grows roughly as $t^{1.5}$. Table I.3 and figure I.2 carry the evidence.

Table I.3: IMU dead reckoning: exponent k of a power-law fit $|e| \propto t^k$ to the anchored position error after the first 5 s (with its R^2), the final heading error, and the share of the final position error that integrating the true velocity through the heading error alone reproduces. A share near one means the position error is heading error integrated along the path; a share well below one means the velocity (accelerometer) integration itself is wrong.

Recording	k	R^2	final error [m]	final heading [°]	heading-only share
`allsensor_000`	1.51	0.94	47.6	2.2	0.01
`allsensor_001`	1.12	0.74	23.7	1.3	0.01
`allsensor_002`	1.54	0.91	17.7	1.4	0.01
`allsensor_003`	1.75	0.93	74.7	-5.8	0.01
`allsensor_004`	1.40	0.87	48.1	-1.4	0.00

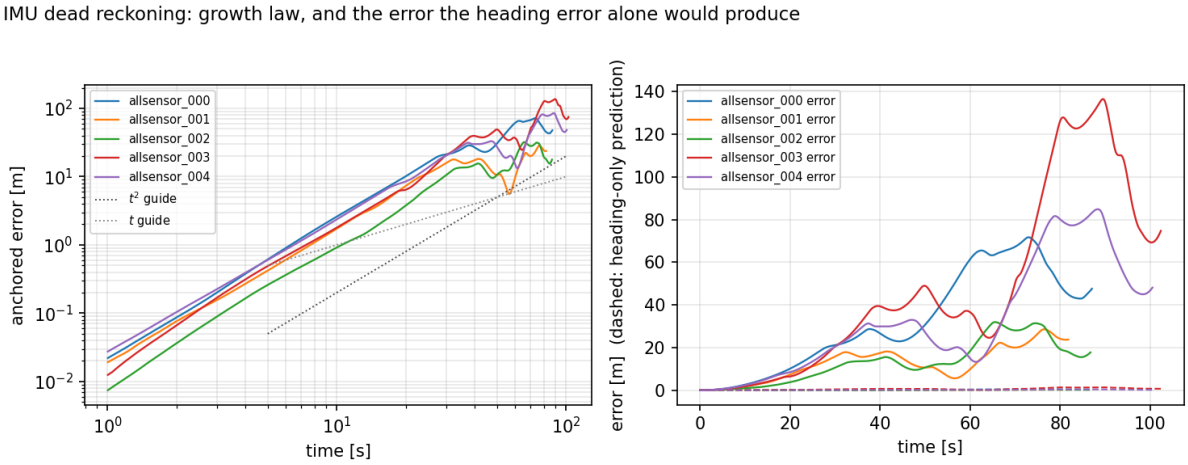


Figure I.2: IMU dead reckoning on the five recordings. Left: anchored position error against time on logarithmic axes, with t and t^2 guides. Right: the same error on linear axes with, dashed, the error that the heading error alone would produce.

The heading error of the IMU integration is 0.6–3.2° RMS, smaller than wheel odometry's, and explains at most one percent of the final position error ($n = 5$). The reported path length is wrong by 1–249 %: the accelerometer integral drifts in speed, and the position error follows a power law with exponent 1.1–1.75 (median 1.5, R^2 0.74–0.94), between the linear growth of a velocity offset and the quadratic growth of an acceleration offset. The gyroscope is not the weak sensor in this platform; the accelerometer integration is. The growth is not concentrated in turns: on all five recordings the position error grows faster on straights, where the speed error accumulates over distance.

4 Insight 3: the EKF removes the heading error, and its covariance stops describing the error at the first turn

The EKF reduces the heading error to 0.3–3.5° RMS and the position error by an order of magnitude, and its NEES leaves the consistent band within one second of the first turn on every recording. Table I.4 and figure I.3 carry the timing; table I.1 the decomposition.

Table I.4: When the EKF's 3-DOF NEES first stays above the 97.5 % band (9.35) for 5 s, as time since the start and as distance travelled, against the time of the first ground-truth heading change (yaw rate above 10 ° s^{-1}); for the EKF with its measured per-recording noise model and for the same filter on the VSLAM noise file. NEES sampled at 1 Hz from the logged covariance. "never": the band is not left for 5 s in the run.

Recording	first turn	EKF, measured noise			EKF @ VSLAM noise		
	[s]	leaves at [s]	at [m]	median NEES	leaves at [s]	at [m]	median NEES
`allsensor_000`	25.6	26.0	37.4	20.9	28.0	39.7	8.4
`allsensor_001`	9.6	9.0	12.4	81.8	10.0	13.7	10.5
`allsensor_002`	12.2	12.0	15.5	78.6	12.0	15.5	23.7
`allsensor_003`	16.4	16.0	21.5	90.6	17.0	22.9	13.8
`allsensor_004`	15.0	15.0	21.4	54.8	17.0	23.7	38.8

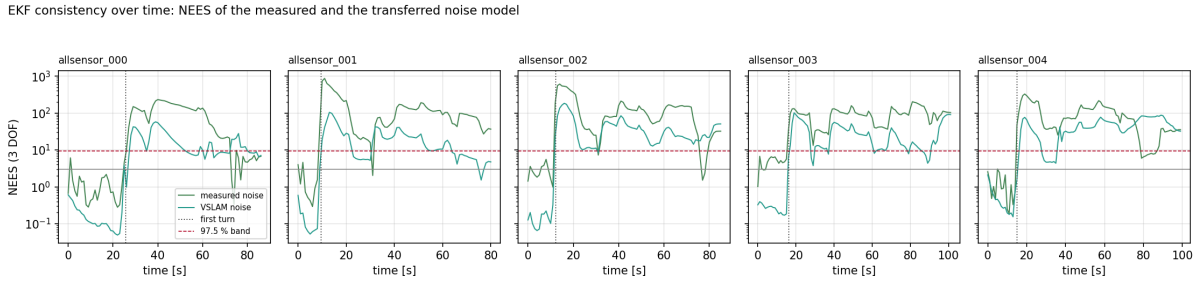


Figure I.3: 3-DOF NEES of the EKF against time on each recording, for the measured per-recording noise model and for the VSLAM noise file, with the first ground-truth heading change (dotted), the 97.5 % band (dashed) and the consistent value of 3 (solid).

Before the first turn the filter is conservative: NEES sits below 3 on every recording. Within 0.6 s of the first heading change it rises by one to two orders of magnitude and never returns ($n = 5$; departure at 9.0–26.0 s against first turns at 9.6–25.6 s). The covariance therefore describes the error on straights and not the heading error that each turn injects: the process model has no term for a turn-dependent heading error, so the state covariance does not grow when the heading does. The two recordings on which the EKF is least accurate, `allsensor_003` and `allsensor_004`, are the two where its heading error shows a slow ramp between the turns of about 0.04–0.07 ° s^{-1} (figure I.1, bottom row), which a gyroscope-bias state should absorb and here does not fully.

[Consistency and calibration are different things] The transferred noise model is less accurate and more consistent: its median NEES is 8–39 against 21–91 for the measured model ($n = 5$), because its larger process noise keeps the covariance closer to the larger error. Neither model is consistent after the first turn. The raw write-up's statement that the calibrated filter is "overconfident by two orders of magnitude" is a whole-run average that mixes a consistent straight with an inconsistent remainder.

5 Insight 4: the transferred noise model turns the EKF back into wheel odometry at the turns

With the VSLAM file's IMU constants the EKF accrues heading error at the turns exactly as the raw wheel odometry does, and that is the whole of the damage the noise transfer causes. The "EKF @ VSLAM noise" rows of tables I.1 and I.2 carry the evidence.

On the transferred model the heading error grows at 431–774 m° s^{-1} while turning and at or below zero on straights, in five of five recordings; the position error grows at 59–130 mm s^{-1} in the turns and at –21–8 mm s^{-1} between them. The signature is the one of section 2: the loosened gyroscope noise density makes the filter weight the wheel yaw rate over the gyroscope in the turns, and the wheel yaw rate is the quantity that is wrong there. The raw write-up's ablation found the gyroscope density to be the responsible

term; this is the mechanism by which it acts. Its path-length error of 9–22 % is a secondary effect of the same rotation: the along-track axis of the path frame moves with the true heading, so a heading error puts part of the lateral error on it.

6 Insight 5: the visual estimators' reported error is largely vertical, and their one failure is a loop-closure event

Scored in the plane, the visual estimators are better than the chapter's three-dimensional ATE says, in four of fifteen runs by a factor of five or more; their planar error does not grow at turns or with fewer features in any consistent way; and the one gross failure is a single 30 m relocalisation at 88 s with the loop closer on, on a recording the same estimator tracks to 0.21 m with it off. Table I.5 and table I.6 and figure I.4 carry the evidence.

Table I.5: Visual estimators scored at the IMU point after the initialisation transient: whole-run aligned ATE in 3-D (as the chapter reports it) and in the plane, the vertical RMS error that separates them, length of the tracking regime, drift rate while turning and on straights in that regime, Spearman ρ between the matched-feature count (ORB-SLAM3 map inliers, VINS-Fusion tracked features) and the drift rate in 2 s bins, and the number of impossible-speed jumps of the estimate.

Recording	Estimator	ATE 3-D [m]	ATE planar [m]	vertical RMS [m]	regime [s]	turning [mm s^{-1}]	straight	ρ (feat., drift) (bins)	jumps
`allsensor_000`	VINS-Fusion	0.56	0.11	0.54	59	10.7	0.6	0.05 (29)	0
	ORB-SLAM3	0.51	0.06	0.50	59	-5.3	2.0	0.08 (29)	1
	ORB-SLAM3 (no LC)	0.04	0.04	0.01	59	-0.4	0.6	-0.03 (29)	2
`allsensor_001`	VINS-Fusion	0.13	0.07	0.10	69	0.4	2.8	-0.11 (34)	0
	ORB-SLAM3	0.10	0.08	0.06	66	3.9	-0.5	0.23 (32)	4
	ORB-SLAM3 (no LC)	0.30	0.29	0.03	66	4.8	0.0	0.11 (32)	3
`allsensor_002`	VINS-Fusion	0.45	0.43	0.12	72	14.2	17.9	0.00 (36)	1
	ORB-SLAM3	0.55	0.53	0.14	72	44.3	-1.8	0.19 (36)	2
	ORB-SLAM3 (no LC)	1.79	1.78	0.13	2	—	—	— (0)	3
`allsensor_003`	VINS-Fusion	0.74	0.71	0.21	84	-9.6	30.0	0.16 (41)	0
	ORB-SLAM3	6.44	6.43	0.31	70	20.6	1.1	-0.07 (34)	6
	ORB-SLAM3 (no LC)	0.22	0.21	0.07	84	-7.0	2.9	0.26 (41)	2
`allsensor_004`	VINS-Fusion	0.28	0.21	0.17	83	-0.4	5.4	-0.16 (41)	0
	ORB-SLAM3	1.17	0.22	1.15	9	—	—	— (0)	3
	ORB-SLAM3 (no LC)	0.76	0.14	0.74	83	-3.0	7.7	-0.14 (41)	3

Table I.6: Anatomy of the ORB-SLAM3 (loop closer on) run on `allsensor_003`: whole-run aligned ATE, the number and sizes of impossible-speed jumps, when they occur, how much of the net error change is taken in jump steps against ordinary drift, and the tracking-state losses the front end logged.

Aligned ATE [m]	6.37
Jumps	6
Jump times [s]	2.4, 2.5, 16.0, 88.3, 89.4, 89.8
Jump sizes [m]	0.41, 0.67, 0.82, 29.99, 11.58, 2.52
Net error change in jump steps [m]	13.60
Net error change in ordinary steps [m]	-2.94
Frames not tracking OK	0
Route progress at the largest jump	0.89
Time of the largest jump [s]	88.3

The vertical component dominates where the three-dimensional ATE is large on a run that tracked: VINS-Fusion on `allsensor_000` is 0.557 m in three dimensions and 0.115 m in the plane (vertical RMS 0.545 m); ORB-SLAM3 on the same recording 0.507 m against 0.060 m; ORB-SLAM3 on

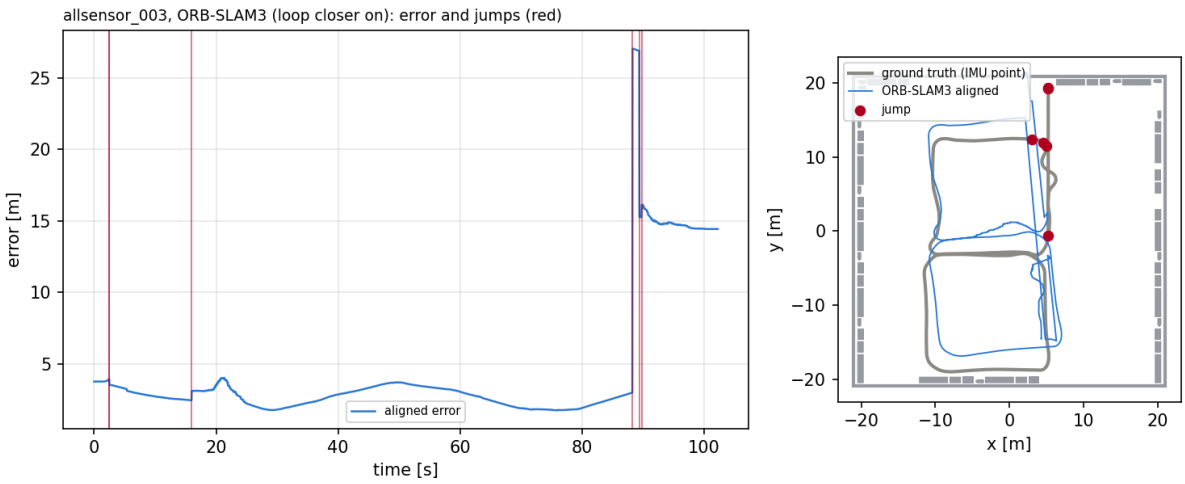


Figure I.4: ORB-SLAM3 with the loop closer on `allsensor_003`. Left: whole-run aligned planar error against time, with the impossible-speed jumps of the estimate in red. Right: the aligned estimate against the ground truth at the IMU point on the warehouse plan, with the jump locations.

`allsensor_004` 1.171 m against 0.219 m, and without the loop closer 0.756 m against 0.135 m. The robot never leaves the floor, so the vertical drift of a visual-inertial estimate is an error the navigation task does not see, while the proprioceptive filter is planar by construction and pays no such penalty. The chapter's estimator comparison therefore charges the visual estimators for a component the baselines cannot have.

[Consequence for the Experiment 1 ranking] In the plane the EKF is the most accurate estimator on one recording of five (`allsensor_002`) and the margins of the other four widen: on `allsensor_000` every visual configuration is below 0.12 m against the filter's 0.19 m, on `allsensor_001` both estimators are below 0.09 m against 0.31 m. Which ATE the chapter quotes for a floor robot is a decision the methodology has to state; the planar value is the one that matches the task.

Turning and feature count do not organise the planar drift. VINS-Fusion drifts faster while turning on two recordings of five and ORB-SLAM3 on three of four with a usable regime; the Spearman correlation between matched features and drift rate is between -0.16 and 0.26 on every run. With one run per estimator per recording neither is a pattern.

The failure is one event. ORB-SLAM3 with the loop closer on `allsensor_003` tracks for 88 s with no frame outside the OK state and a planar error below 4 m under the whole-run fit (the fit itself is pulled by what follows), then takes jumps of 30.0, 11.6 and 2.5 m within 1.5 s. The net error change inside the jump steps is 13.6 m; outside them it is −2.9 m. The run without the loop closer on the same recording ends at 0.21 m. The event is therefore a loop closure or map merge that accepted a wrong correspondence, late in the route, not tracking loss and not drift; the raw write-up's flag on this value can be replaced by that attribution.

7 What these data cannot answer

- **Why the per-turn heading error of wheel odometry varies in size.** It is not the fitted yaw-rate scale alone (H1 of the raw write-up); slip at the steering limit, the Ackermann model's track width and the replay rate are candidates none of the logs isolate.
- **Whether the EKF's residual heading ramp is gyroscope bias.** The bias state is logged in variance only, not in value, so the ramp on `allsensor_003` and `004` is attributed to it by elimination.
- **Whether the loop-closure failure repeats.** One run with the loop closer on exists per recording;

ORB-SLAM3 is non-deterministic.

- **Anything about turning or features for the visual estimators with confidence.** Five recordings, one run each.

8 Summary

Table I.7: Questions, answers and the strength of the evidence.

	Answer	Evidence	Confidence
Q1 wheel error	Heading error carried into position; distance within 1 %	heading-only share 0.95–1.04 on 5/5	high
Q2 turning	Wheel and transferred-noise EKF: heading made in turns, 5/5; IMU: no; visual: no pattern	table I.2	high / high / low
Q3 IMU divergence	Velocity integral, exponent ≈ 1.5; heading explains ≤ 1 %	table I.3	high
Q4 EKF consistency	Consistent until the first turn, inconsistent from it, 5/5 within 1 s	table I.4	high
Q5 noise transfer	Re-weights the wheel yaw rate in turns; heading error at turns like raw wheel odometry	"EKF @ VSLAM noise" rows	high
Q6 visual error	Vertical drift dominates 4/15 three-dimensional ATEs; planar error 0.04–0.7 m; one loop-closure event	tables I.5 and I.6	high (composition), medium (event)

Consequences for Chapter 4. State which ATE a floor robot is scored on and quote the planar value beside the three-dimensional one for the visual estimators. Report the EKF's consistency as "until the first turn" rather than as a whole-run ANEES. Replace H1's falsification with the mechanism: per-turn heading error of varying size. Attribute the ORB-SLAM3 6.4 m value to a single late loop-closure event rather than flagging it as unexplained. Name the accelerometer integral, not the gyroscope, as the reason IMU-only dead reckoning is unusable, since that is what a VIO fusion is being asked to fix.